- **Internship Name : Data Analysis**
- **Task Name: Task 7: Learn Pandas Series and DataFrame basics: creating a DataFrame, viewing data with head(), info(), describe().**
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- **Internship Organization :**
- **Date:**

1.Introduction:

Pandas is an open-source Python library used for data analysis and manipulation. It provides two main data structures, **Series** (one-dimensional) and **DataFrame** (two-dimensional, table-like). In this task, I worked with these structures and used basic functions to explore and understand a dataset.

2. Objectives:

- To understand and create a Pandas Series
- To create a Pandas DataFrame from a dictionary
- To view data using head()
- To understand the dataset structure using info()
- To get statistical summaries using describe()

3. Tools and Technologies Used:

- **Language:** Python 3.13
- **Library:** Pandas
- **Environment:** Jupyter Notebook

4. Methodology and Implementation:

Step 1: Importing the Library

I imported the Pandas library using import pandas as pd.

```
import pandas as pd
```

Step 2: Creating a Pandas Series

A Series is a one-dimensional labeled array. I created a Series of student marks: [75, 82, 68, 90, 85]. Pandas automatically assigned index values 0 to 4, and the data type was int64.



Step 3: Creating a Pandas DataFrame

A DataFrame is a two-dimensional table with rows and columns. I created one from a Python dictionary containing student records with four columns:

Column	Description
Name	Student name
Age	Student age
Marks	Marks obtained
City	City of the student

The DataFrame had 5 records (Amit, Priya, Rahul, Sneha, Neha).

```
##Step 3 – Pandas DataFrame
```

```
import pandas as pd

data = {
    "Name": ["Amit","Priya","Rahul","Sneha","Neha"],
    "Age": [22,23,24,21,25],
    "Marks": [75,82,68,90,85],
    "City":["Nagpur","Pune","Mumbai","Nagpur","Pune"]
}

df = pd.DataFrame(data)
print(df)
```

✓ 0.8s

```
    Name  Age  Marks    City
0   Amit   22     75  Nagpur
1  Priya   23     82    Pune
2  Rahul   24     68  Mumbai
3  Sneha   21     90  Nagpur
4   Neha   25     85    Pune
```

Step 4: Using the head() Function:

- df.head() displays the first 5 rows by default.
- df.head(3) displays only the first 3 rows.

This is useful for quickly previewing a large dataset without printing everything.

```
##Step 4 – head() Function
```

```
df.head()
```

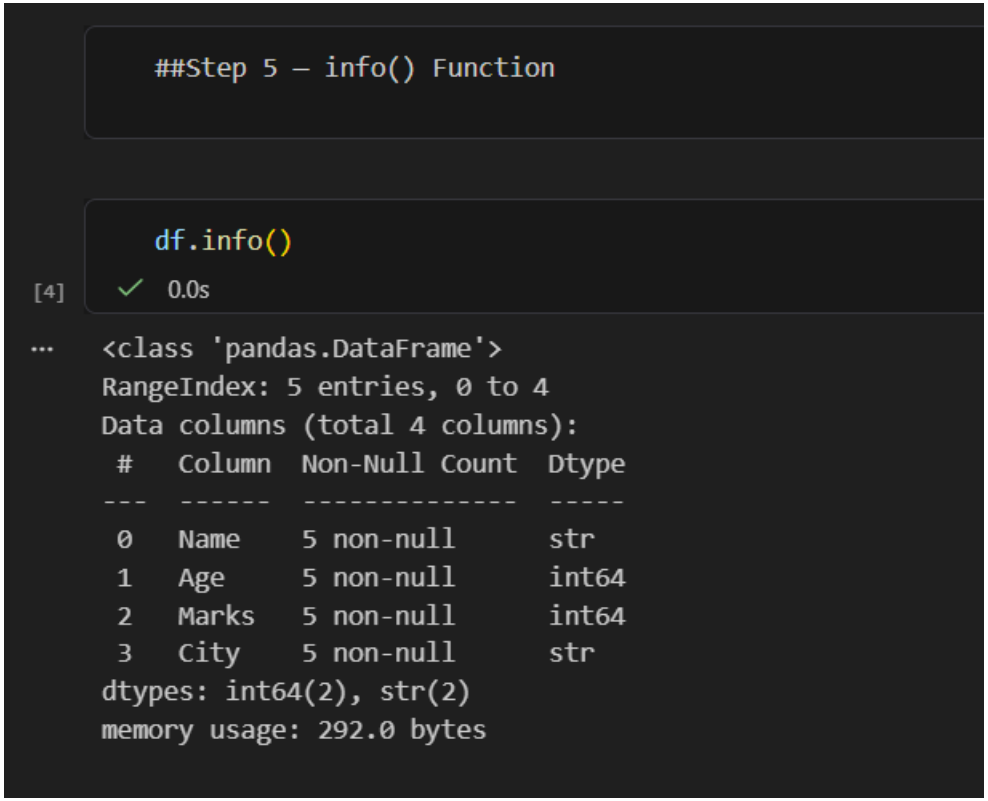
✓ 0.0s

	Name	Age	Marks	City
0	Amit	22	75	Nagpur
1	Priya	23	82	Pune
2	Rahul	24	68	Mumbai
3	Sneha	21	90	Nagpur
4	Neha	25	85	Pune

Step 5: Using the info() Function

df.info() gives a concise summary of the DataFrame. The output showed:

- The DataFrame has **5 entries** (RangeIndex 0 to 4) and **4 columns**
- **No missing values** (all columns have 5 non-null entries)
- Data types: Age and Marks are int64; Name and City are string type
- Memory usage: 292 bytes



Step 6: Using the describe() Function

For numerical columns (df.describe()):

```
##Step 6 – describe() Function
```

```
df.describe()
```

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	Age	Marks
count	5.000000	5.000000
mean	23.000000	80.000000
std	1.581139	8.631338
min	21.000000	68.000000
25%	22.000000	75.000000
50%	23.000000	82.000000
75%	24.000000	85.000000
max	25.000000	90.000000

- **Name:** 5 unique values, each appearing once
- **City:** 3 unique values (Nagpur, Pune, Mumbai); the most frequent is **Nagpur** (2 times)

5. Observations:

- The average marks of the students is **80**, and the average age is **23** years.
- The highest marks are **90** (Sneha) and the lowest are **68** (Rahul).
- The dataset has no missing values, so no data cleaning was required.
- Nagpur and Pune each have 2 students, and Mumbai has 1.
- A deprecation warning appeared when using include="object" in the newer Pandas version, because text columns now use the str dtype. In future versions, include="str" should be used instead.

6. Challenges Faced:

- Understanding the difference between a Series and a DataFrame
- Handling the Pandas deprecation warning for describe(include="object")

7. Learning Outcomes:

- I learned how to create and work with Pandas Series and DataFrames.
- I learned how to quickly inspect a dataset using head(), info() and describe().
- I understood how to read basic statistics such as mean, median, standard deviation, and quartiles.
- I gained a foundation for data analysis, which is useful for further tasks such as data cleaning, filtering, and visualization.

8. Conclusion:

This task gave me practical, hands-on experience with the fundamentals of Pandas. By creating a Series and a DataFrame and applying head(), info() and describe(), I learned how to load, inspect, and summarize data. These skills are the first step in any data analysis workflow.